

User-Centric Definitions of Quality Content and Engagement on Instagram

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Abstract

This intervention examines young adults' subjective and heterogeneous definitions of quality content, moving beyond objective or academic-driven definitions. Using a two-wave, user-centric intervention design, 65 participants aged 18 to 29 donated Instagram data and provided open-ended definitions of quality content. In Wave 1, participants' responses were thematically analyzed and synthesized into a crowd-based definition, alongside donated Likes data. In Wave 2, participants donated additional data, enabling comparisons between user-centric, crowd-based, and academic definitions of quality content. Qualitative analysis identified five dominant categories: mood-enhancing, educational, authentic, socially interactive, and entertainment content. Quantitative analyses examined the alignment between these definitions and participants' actual engagement behavior. Results reveal substantial interindividual variation, indicating that perceptions of quality content are highly personalized and context-dependent.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Quality content, data donation, intervention, user-centric approach, automated analysis, panel survey, mood

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1 Introduction

Visual social media platforms such as Instagram have become integral to young adult's everyday lives, offering spaces for interactions, self-expression, and information exchange. The platform exposes users to an immense variety of content—ranging from entertainment and lifestyle to news and social causes—making them central to how individuals form opinions, seek inspiration, and construct meaning online. While this abundance of content presents unique opportunities for learning and connection, it also raises critical questions about what constitutes “quality” content and how users themselves define and experience it [9].

Despite growing academic and public interest in media quality, most research has relied on externally imposed definitions—such as informational accuracy, prosociality, or production value—largely overlooking user's subjective understandings of content itself [3, 7]. As a result, existing literature often does not capture the diversity and nuance in how people evaluate and engage with content in their daily social media environments. Moreover, studies frequently depend on general measures of social media use (e.g., time spent online or frequency of posting), which obscure the heterogeneity of content experiences and their potential implications for well-being [10].

Recent efforts to refine measurement approaches have focused on content categories such as exposure to positive content and posting positive content, such as travel or appearance-related posts [8] or authenticity related liking activities [9]. However, such predefined categories may not reflect users' own conceptualizations of meaningful or beneficial media experiences. A user-centric approach is therefore needed to understand how individuals themselves define quality content and how these definitions translate into actual engagement behavior [4].

The present study addresses this gap by examining how young adults (aged 18–29) define and interact with quality content on Instagram. Using a data donation approach combined with self-reported user definitions, we investigate (1) how young adults conceptualize quality content in their own words, (2) how these user-centric definitions compare to an aggregated crowd definition and more abstract academic definition, and (3) whether their content engagement patterns on Instagram align with these definitions.



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Our novel methodological approach aims to understand how young adults define and engage with quality content online. We administered a quality content intervention with a particular focus on heterogeneity—that is, on how one person’s definition of quality content may differ greatly from the other. Building on pre-established academic definitions of quality content in media, the study was designed to capture user-centric definitions and examine their variation across individuals. Based on these user-provided definitions, we developed an aggregated “crowd definition” of quality content. Through this multi-layered analysis, the study contributes to a deeper understanding of (1) how active Instagram users perceive and make sense of the content on their feeds, (2) how they define and evaluate quality content for themselves, and (3) how these definitions correspond to their real engagement behaviors on Instagram.

Within media and communication research, quality content has often been defined through theoretically grounded frameworks emphasizing its cognitive, emotional, and social value. Building on the E-AIMS theoretical framework [5], quality content can be understood as content which promotes engaged, active, meaningful, and socially interactive experiences. Each of these dimensions captures a distinct but interrelated aspect of how people derive value from digital media. Engaging content captures and sustains attention through coherent narratives, aesthetic appeal, or emotional depth. Active content stimulates cognitive or behavioral involvement—encouraging users to think, comment, create, or reflect rather than passively consume. Meaningful content connects to users’ everyday lives, experiences, and identities, allowing them to find personal relevance in the messages they encounter. Finally, socially interactive content invites collaboration, communication, and collective meaning-making through social affordances such as tagging, commenting, or sharing.

These definitions provide structured and generalizable criteria, but may not fully capture how users experience content in everyday, highly personalized social media contexts. A user-centered perspective highlights subjective evaluations of quality, such as whether content feels authentic, uplifting, or socially relevant. Rather than viewing these perspectives as opposing, this study treats them as complementary: academic definitions offer normative guidance, while user-centered definitions reflect lived experience. We focus on Instagram as a visually driven platform where content evaluation is shaped by entertainment, aesthetics, authenticity, and social interaction, making it particularly relevant for studying perceptions of “quality content.” At the same time, we limit our claims to this context, acknowledging that platforms with different affordances (e.g., text-based or discussion-oriented) may produce different patterns of content evaluation and engagement. Prior CHI work has examined content quality and engagement from complementary perspectives. For example, research on Instagram creators highlights how engagement and success are shaped by content features and platform dynamics, often emphasizing metrics such as likes, visibility, and audience reach [11]. In contrast, other CHI studies focus on user perceptions of low- vs. high-quality content, developing taxonomies that show how subjective experiences of content quality can diverge from objective or system-driven measures [12]. Additionally, work on platforms like Twitter demonstrates that

algorithmic or design interventions can unintentionally reduce content quality, underscoring tensions between platform optimization and user experience [6].

Taken together, these elements define quality content not simply as media that entertains or informs, but as content that contributes to learning, empathy, and digital well-being. In contemporary media environments, this academic understanding provides a structured foundation for identifying how social media posts, reels, or stories might fulfill users’ deeper psychological and social needs, beyond surface-level engagement metrics such as likes or views. Previous studies imply changes in well-being due to exposure and content based engagement with unique personalized content on Instagram [1–3], but do not show how personally relevant and meaningful the content is to users. Thus, we ask:

RQ1: What topics do active Instagram users most frequently engage with through liking and are there any gender differences?

RQ2: How do active Instagram users define quality content on the platform?

RQ3: To what extent do the academic, user-centric, and crowd-aggregated definitions of quality content overlap?

RQ4: To what extent do the three quality content definitions relate with positive mood two-weeks later at Wave 2?

2 Study Design and Method

The intervention study goes beyond objective categorizations of quality content, focusing on the heterogeneity and subjective experiences of young adults’ definitions and engagement with quality content. Young adults ($N_{Wave1} = 65$, $N_{Wave2} = 55$; 18–29-year-olds) participated in a two-wave intervention designed to capture individual differences in perceptions and definitions of quality content through a user-centric and data donation approach on Instagram. As part of the pilot study, in the first wave, participants provided open-ended definitions of what quality content means to them, which were thematically analyzed and synthesized into an aggregated “crowd definition.” Additionally, participants donated their data from their *Likes* page on Instagram ($n_{Wave1} = 622$). In the second wave, two-weeks later, participants donated data again ($n_{Wave2} = 554$) resulting in $N = 1176$ single liked Instagram posts operationalized as active engagement, allowing for content comparisons between actual content engagement, individual user-centric definition, crowd definition, and academic definition of quality content. First, we conducted qualitative analysis of user-centric definitions identified dominant user-centric categories. Second, through quantitative analyses we explored the overlap between the user, crowd and academic definitions and participants’ actual liked Instagram posts suggesting some overlap across donated liked posts and the three definitions. Third, we conducted linear mixed model predicting positive mood across two-wave surveys while controlling for key sociodemographics.

3 Results and Discussion

3.1 Frequency of Topics

In the RQ1, we asked what are the topics participants engage with most. Through automated content analysis assigning three topics per each post in one thumbnail, the results revealed most frequent topics. A comparison of the most frequent liked topics of liked

Instagram content by gender. Table 1 presents the top 20 frequent topic words for each gender, revealing both similarities and differences. Both female ($n = 34$) and male users ($n = 9$) ranked topics such as “Outdoor,” “Friends,” and “Education” highly, indicating shared interests in social connection and learning environments. However, the topic analysis revealed distinct patterns in content preferences by gender. Females tended to use more interpersonal and lifestyle-related words such as “friends,” “fashion,” and “celebration,” suggesting a focus on social connection and self-presentation. Males, in contrast, used more activity-oriented and competitive terms like “sports,” “team,” and “competition,” reflecting interests in group activities and performance. These findings highlight how gendered differences in topic frequency may reflect broader social and cultural patterns in content consumption, however, due to small and gender-imbalanced sample should be interpreted with caution.

Wave-specific patterns also emerged. In Wave 1, users engaged more frequently with topics such as mental health, beauty, health, technology, animals, and athlete-related content, pointing to a focus on self-expression, wellbeing, and personal achievements. In contrast, Wave 2 saw an increase in more aesthetic and event-oriented topics such as art, black and white photography, ceremony, concert, culture, style, and wedding, reflecting a shift toward curated visual presentation and social celebration. Overall, the results indicate that while the core of user engagement centers around social and lifestyle-oriented themes, there is evidence of temporal variation in the visual and cultural framing of posts across waves.

3.2 User-Centric Definitions of Quality

The RQ2 asked, how do active Instagram users define quality content. Participants’ open-ended definitions of quality content that were analyzed inductively to identify recurring themes and underlying evaluative criteria. Across 54 responses collected at Wave 1, five major categories of quality content emerged. Each category reflects a distinct dimension of how young adults conceptualize value and meaning in social media posts. The overview of the results is presented in Table 2.

3.2.1 Positive and Mood-Enhancing Content. The most frequently mentioned category describes content that evokes positive emotions or enhances mood. Participants emphasized posts that make them “happy,” “laugh,” or “feel inspired.” This type of content often included wellness, motivational, spiritual, and “hope core” accounts. Representative phrases included “things that make me happy,” “lifts my mood,” “makes me laugh,” and “uplifting or inspiring content.”

3.2.2 Informative and Educational Content. The second most common category emphasized the utility and learning value of content. Participants valued posts that provide new information or practical knowledge applicable to daily life. Common examples included news updates, health and fitness advice, or life tips. Typical responses included “beneficial to my life,” “helps me learn something new,” “educational and relevant,” and “useful to my everyday life.”

3.2.3 Personal and Authentic Content. The third category reflected a preference for authenticity and real-life expression. Participants described quality content as “real,” “relatable,” and “genuine,” often contrasting it with content perceived as staged or artificial. Examples included personal updates, relatable humor, or creators who

share similar life experiences. Example phrases were “real content, nothing fake,” “personal life updates,” and “relatable and looks like the people are having fun.”

3.2.4 Social Connection and Relationship-Oriented Content. Participants also defined quality content in terms of connection—posts that strengthen social bonds or maintain awareness of others’ lives. This included updates from friends, family, or individuals with shared backgrounds. Illustrative phrases included “life updates from connections,” “close friends,” “helps me feel connected,” and “updates on my friends’ lives.”

3.2.5 Creative, Aesthetic, and Entertaining Content. Finally, participants valued content for its creativity and visual appeal. Posts described as “aesthetic,” “entertaining,” or “trend-based” were cited as forms of quality content. Examples included dance videos, fashion posts, music clips, and other forms of visually engaging or trend-driven media. Example phrases included “aesthetic content,” “cool shots,” “entertaining and funny,” and “trends, popular culture.”

3.3 Overlaps Between Academic, Crowd, and User-Centric Definitions

As an answer to RQ3, we examined the extent to which the academic, crowd, and user-centric definitions of quality content overlap. The descriptive statistics revealed clear trends in how quality content was characterized across definitions. Under the academic definition, active content appeared in all liked posts ($n = 1,142$), confirming that each post involved user activity. Meaningful content was also frequent ($M = 0.74$, $SD = 0.44$; $n = 842$), followed by interactive content ($M = 0.46$, $SD = 0.50$; $n = 520$) and engaged content ($M = 0.29$, $SD = 0.45$; $n = 327$). Within the crowd definition, positive posts were the most common ($M = 0.83$, $SD = 0.37$; $n = 950$), indicating that the majority of content was emotionally uplifting. Authentic content followed ($M = 0.59$, $SD = 0.49$; $n = 674$), while entertainment ($M = 0.36$, $SD = 0.48$; $n = 410$), informative ($M = 0.24$, $SD = 0.43$; $n = 276$), and social ($M = 0.23$, $SD = 0.42$; $n = 267$) content were less frequent. According to the user definition, posts categorized as user-defined quality content appeared in a substantial proportion of cases ($M = 0.68$, $SD = 0.47$; $n = 775$). The findings suggest that content perceived as positive, meaningful, and authentic dominated across the data set, whereas social and informative elements were comparatively limited.

Next, we conducted correlational analyses between summed quality content indicators. Table 3 shows significant positive associations between all three definitions of quality content. The academic and crowd definitions were weakly but significantly correlated ($r = .13$, $p < .001$), indicating a small degree of overlap between theoretically defined and crowd-perceived indicators of content quality. The user definition correlated moderately with both the academic ($r = .27$, $p < .001$) and crowd ($r = .22$, $p < .001$) definitions, suggesting that users’ own perceptions of what makes content “good” align somewhat with both expert (academic) and community (crowd) views, though the associations are not strong. Overall, these results suggest that although the three conceptualizations of quality content share some common ground, the academic, crowd-aggregated, and user-centric definitions show limited conceptual and empirical overlap, reflecting largely distinct dimensions of perceived content

Table 1: Top 20 Frequent Topic Words for Each Gender in Descending Order

Number Order	Female	Male	Number Order	Female	Male
1	Portrait	Sports	11	Event	Music
2	Outdoor	Friends	12	Celebration	Nature
3	Friends	Lifestyle	13	Performance	Fashion
4	Fashion	Outdoor	14	Party	Group
5	Sports	Portrait	15	Dance	Outdoors
6	Lifestyle	Tennis	16	Social	Fitness
7	Graduation	Social	17	Music	Team
8	Education	Education	18	Travel	Competition
9	Nature	Graduation	19	Food	Party
10	Group	Family	20	Selfie	Athlete

Table 2: Overview of Quality Content Categories Based on Participants' Definitions

Category	Description	Mentions (n)	Percentage (%)
Positive / Mood-Enhancing Content	Content that makes people happy, laugh, or feel inspired; includes wellness, motivational, and "hope core" posts.	17	34
Informative / Educational Content	Content that teaches something new or is practically relevant; includes news, health, and lifestyle advice.	14	28
Personal / Authentic Content	Content that feels genuine and real rather than staged; includes personal updates and relatable humor.	11	22
Social Connection / Family and Friends Updates Content	Content that maintains social ties and relational awareness; includes updates from friends and peers.	9	18
Creative / Entertaining Content	Visually appealing or trend-driven posts; includes art, music, fashion, and pop culture content.	8	16

Note. $N = 54$. Percentages are approximate and based on 54 open-ended responses provided at Wave 1.

Table 3: Correlations Between Definitions

Variables	Academic Def.	Crowd Def.
Academic Definition Sum	—	
Crowd Definition Sum	.13***	—
User Definition	.27***	.22***

Note. $N = 1,142$. Pearson correlation coefficients and p-values (two-tailed) between the summed indices of academic, crowd, and user definitions of quality content.

quality. In terms of overlap, frequency analysis of summed indicators showed the largest overlap ($n = 774$) between the academic and crowd definitions, suggesting moderate convergence between theoretically grounded and crowd-based definitions of quality. In contrast, the user-centric definition showed almost no overlap with either academic or crowd definitions ($n = 1$), indicating that participants' subjective perceptions of quality diverged substantially from both academic and crowd definitions.

3.4 Quality Content Predicting Positive Mood

Answering RQ4, a linear mixed-effects model presented in Table 4 was estimated to examine whether academic, crowd, and user-defined indices of quality content predicted participants' positive mood while controlling for age, gender, and wave 1 baseline measure. The model included a random intercept for participant ID to account for repeated measures within individuals. The analysis was based on 1,112 observations nested within 39 participants. The

model converged successfully ($\text{REML} = 573.6$). The random effects indicated substantial between-person variability in baseline positive mood (variance = 0.55, $SD = 0.74$), and smaller within-person residual variance (variance = 0.08, $SD = 0.28$). These results suggest that most of the variance in positive mood was attributable to stable individual differences rather than situational fluctuations. As shown in Table 4, none of the content quality predictors were significant. Academic content quality ($B = -0.00$, $SE = 0.01$, $t = -0.35$), crowd-rated quality ($B = -0.02$, $SE = 0.01$, $t = -1.84$), and user-defined quality ($B = -0.01$, $SE = 0.02$, $t = -0.23$) were unrelated to positive mood. Similarly, age ($B = 0.04$, $SE = 0.06$, $t = 0.69$) and gender ($B = -0.07$, $SE = 0.29$, $t = -0.24$) did not significantly predict mood. The only significant effect was wave ($B = -0.13$, $SE = 0.02$, $t = -6.50^*$), indicating that the positive mood declined across successive measurement periods. The results indicate that neither theoretically derived (academic), socially aggregated (crowd), nor self-defined (user) indices of quality content were associated with participants' positive mood. The decline in mood over time suggests a temporal or fatigue-related effect rather than one linked to content engagement.

4 Discussion

This study highlights that perceptions of "quality content" on Instagram are highly subjective and multidimensional, with users emphasizing mood-enhancing, educational, authentic, social, and entertaining content. While there is some overlap between academic, crowd-based, and user-defined definitions, the relatively

Table 4: Linear Mixed Model Predicting Positive Mood at Wave 2

Predictor	Estimate (b)	SE	t value
(Intercept)	2.89	1.25	2.31
Academic Definition	-0.00	0.01	-0.35
Crowd Definition	-0.02	0.01	-1.84
User Definition	-0.01	0.02	-0.23
Age	0.04	0.06	0.69
Gender	-0.07	0.29	-0.24
Wave 1	-0.13	0.02	-6.50***

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

weak associations suggest they capture distinct aspects of content quality. Although users tend to engage with positive and meaningful content, these quality indicators did not predict positive mood over time, pointing to the importance of contextual and individual factors beyond content characteristics [11]. Observed variation in preferences, including gender differences and shifts across waves, further underscores the dynamic and personalized nature of content engagement. From an HCI perspective, the findings suggest that current platform metrics may not fully reflect what users perceive as valuable. Incorporating user-centered definitions of quality could support more meaningful and personalized content curation.

5 Limitations

The study has several limitations. The small and gender-imbalanced sample limits generalizability due to its pilot study mode and preliminary results. The focus on Instagram restricts applicability to other platforms with different affordances. The data donation approach may introduce selection bias and does not capture all forms of engagement. Additionally, the short time frame limits conclusions about long-term effects on well-being.

6 Conclusion and Future Work

This pilot study offers a preliminary overview of how young adults define and experience quality content on Instagram by comparing user-centered, crowd-based, and academic definitions with their engagement behavior. Findings highlight considerable heterogeneity across users and their definitions and liked posts, suggesting that perceptions of quality content are highly individualized and context-dependent. As data collection continues with a larger youth sample, this work aims to refine these insights and inform future design and platform practices that prioritize user-defined quality over algorithmic curation.

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